

Chapter One: Introduction to MIS

DEFINITION AND IMPORTANCE OF MIS

- ▶ MIS is an information system used by organizations to collect, store, process, and distribute information for decision-making purposes.
- ▶ MIS systems are typically used to support a variety of business functions, including:
 - ▶ **Financial management:** MIS systems can be used to track financial data, such as revenues, expenses, and profits.
 - ▶ This information can be used to make informed decisions about resource allocation, pricing, and investment.
 - ▶ **Operational management:** MIS systems can be used to track operational data, such as inventory levels, production schedules, and customer orders.
 - ▶ This information can be used to identify and resolve problems, improve efficiency, and optimize resource utilization.
 - ▶ **Marketing management:** MIS systems can be used to track marketing data, such as customer demographics, sales performance, and campaign results.
 - ▶ This information can be used to identify target markets, develop effective marketing campaigns, and measure the effectiveness of marketing efforts.

DEFINITION AND IMPORTANCE OF MIS

- ▶ **Human resources management:** MIS systems can be used to track human resources data, such as employee records, payroll information, and training records.
- ▶ This information can be used to manage employee performance, make hiring and promotion decisions, and comply with labor laws.
- ▶ *MIS systems are typically composed of three main components:*
- ▶ **Hardware:** The physical components of the system, such as computers, servers, and storage devices.
- ▶ **Software:** The programs that run on the hardware, such as databases, spreadsheets, and reporting tools.
- ▶ **Data:** The information that is collected, stored, and processed by the system.

DEFINITION AND IMPORTANCE OF MIS

- ▶ MIS systems can be developed in-house or purchased from a vendor.
- ▶ MIS software products available, ranging from general-purpose systems to specialized industry-specific solutions.
- ▶ Here are some of the key reasons why MIS is so important:
- ▶ ***Improved decision-making***
- ▶ ***Increased efficiency and productivity***
- ▶ ***Reduced costs***
- ▶ ***Enhanced collaboration and communication***
- ▶ ***Improved customer service***
- ▶ ***Competitive advantage***

DEFINITION AND IMPORTANCE OF MIS

- ▶ Healthcare organizations can use MIS to manage patient records, track treatment outcomes, and improve patient care.
- ▶ Manufacturing companies can use MIS to monitor production processes, optimize supply chains, and ensure product quality.
- ▶ Retail businesses can use MIS to analyze customer purchasing patterns, develop targeted marketing campaigns, and manage inventory levels.

- ▶ composed of several subsystems that work together to collect, store, process, and distribute information for decision-making purposes.
- ▶ These subsystems can be categorized into three main groups:
- ▶ **Transaction Processing Systems (TPS)**: responsible for capturing and recording routine business transactions, such as sales orders, inventory movements, and payroll transactions.
- ▶ Involve high-volume, repetitive tasks and are often integrated with other systems, such as inventory management systems and accounting systems.

SUBSYSTEMS OF MIS

▶ ***Management Reporting Systems (MRS)***

- ▶ Designed to generate reports and summaries of data collected from TPS and other sources.
- ▶ Provide managers with the information they need to track performance, identify trends, and make informed decisions.

▶ ***Decision Support Systems (DSS)***

- ▶ Provide managers with the information and tools they need to make complex decisions.
- ▶ Use data from TPS and MRS, as well as other sources, such as external databases and market research reports.

SUBSYSTEMS OF MIS

- ▶ In addition to these three main categories, there are a number of other specialized MIS subsystems
- ▶ ***Executive Information Systems (EIS)***
- ▶ ***Office Information Systems (OIS)***
- ▶ ***Knowledge Management Systems (KMS)***

CONTEMPORARY APPROACHES TO MIS

- ▶ Refer to the current methodologies, frameworks, and technologies used to design, implement, and manage information systems.
- ▶ Strategic Alignment
- ▶ Business Process Orientation
- ▶ Emergent Technologies
- ▶ Human-Centered Design
- ▶ Information Governance and Risk Management
- ▶ ***Examples of Contemporary MIS Approaches:***
 - ▶ Enterprise Resource Planning (ERP)
 - ▶ Business Intelligence (BI)

CONTEMPORARY APPROACHES TO MIS

- ▶ Knowledge Management (KM)
- ▶ Social Media Analytics
- ▶ Mobile Computing

- ▶ The evolution of the information age can be broadly divided into three phases:
- ▶ **1. Pre-Information Age (Before the Mid-20th Century)**
- ▶ information was primarily shared through oral traditions, written texts, and limited forms of communication like the printing press.
- ▶ Access to information was restricted and largely confined to the elite and educated classes.
- ▶ **2. Early Information Age (Mid-20th Century to Mid-1970s)**
- ▶ marked the transition to a more digitized and interconnected world.
- ▶ The invention of the transistor, the development of mainframe computers, and the rise of telecommunications laid the foundation for the information revolution.

▶ **3. *Digital Information Age (Mid-1970s to Present)***

- ▶ Rapid advancements in computing power.
- ▶ Widespread adoption of personal computers and the internet, and
- ▶ Emergence of new technologies like mobile computing, cloud computing, and artificial intelligence.
- ▶ ***Key Milestones in the Evolution of the Information Age:***
 - ▶ **1947:** Invention of the transistor
 - ▶ **1951:** First stored-program computer
 - ▶ **1969:** ARPANET, the precursor to the internet, established.

- ▶ **1971:** Microprocessor, a complete computer on a single chip, invented by Intel.
- ▶ **1973:** Robert Cailliau and Tim Berners-Lee develop the concept of the World Wide Web.
- ▶ **1981:** IBM introduces the personal computer
- ▶ **1990s:** Rise of the internet, email, and web browsers
- ▶ **2000s:** Emergence of mobile computing, cloud computing, and social media.
- ▶ **2010s and beyond:** Continued advancements in artificial intelligence, big data analytics, and internet of things technologies, shaping the future of the information age.

- ▶ **IT** is the study, design, development, implementation, support or management of computer-based information systems.
- ▶ encompasses a broad range of hardware and software, as well as the people and processes.
- ▶ key areas of IT:
- ▶ **Hardware:** This includes the physical components of computer systems, such as processors, memory, storage devices, and input/output devices.
- ▶ **Software:** This includes the programs that run on computers, such as operating systems, application software, and middleware.
- ▶ **Networking:** This includes the technologies that allow computers to communicate with each other, such as the Internet and local area networks (LANs).
- ▶ **Data management:** This includes the processes and technologies used to collect, store, organize, and retrieve data.
- ▶ **Security:** This includes the technologies and practices used to protect information from unauthorized access, use, disclosure, disruption, modification, or destruction.

INFORMATION TECHNOLOGY AND RELATED CONCEPTS

- ▶ **Most common IT careers:**
- ▶ **IT project manager:** IT project managers plan, organize, and execute IT projects.
- ▶ **Software developer:** Software developer design, code, and test software applications.
- ▶ **Network administrator:** Network administrators install, configure, and maintain computer networks.
- ▶ **Database administrator:** responsible for ensuring that databases are secure, reliable, and available to users.
- ▶ **Information security analyst:** protect information systems from unauthorized access, use, disclosure, disruption, modification, or destruction.

Chapter Two: Foundational Concept in MIS

- ▶ Broadly categorized into four main areas.
- ▶ **Planning** involves setting goals and objectives, developing strategies to achieve those goals.
- ▶ Determining what needs to be done, when it needs to be done, and how it will be done.
- ▶ **Organizing** involves structuring the organization, assigning tasks and responsibilities, and establishing communication channels.
- ▶ **Leading** involves motivating and inspiring employees, providing direction and guidance.
- ▶ **Controlling** involves monitoring progress, evaluating performance, and taking corrective actions as needed.

Business and Management Functions

18

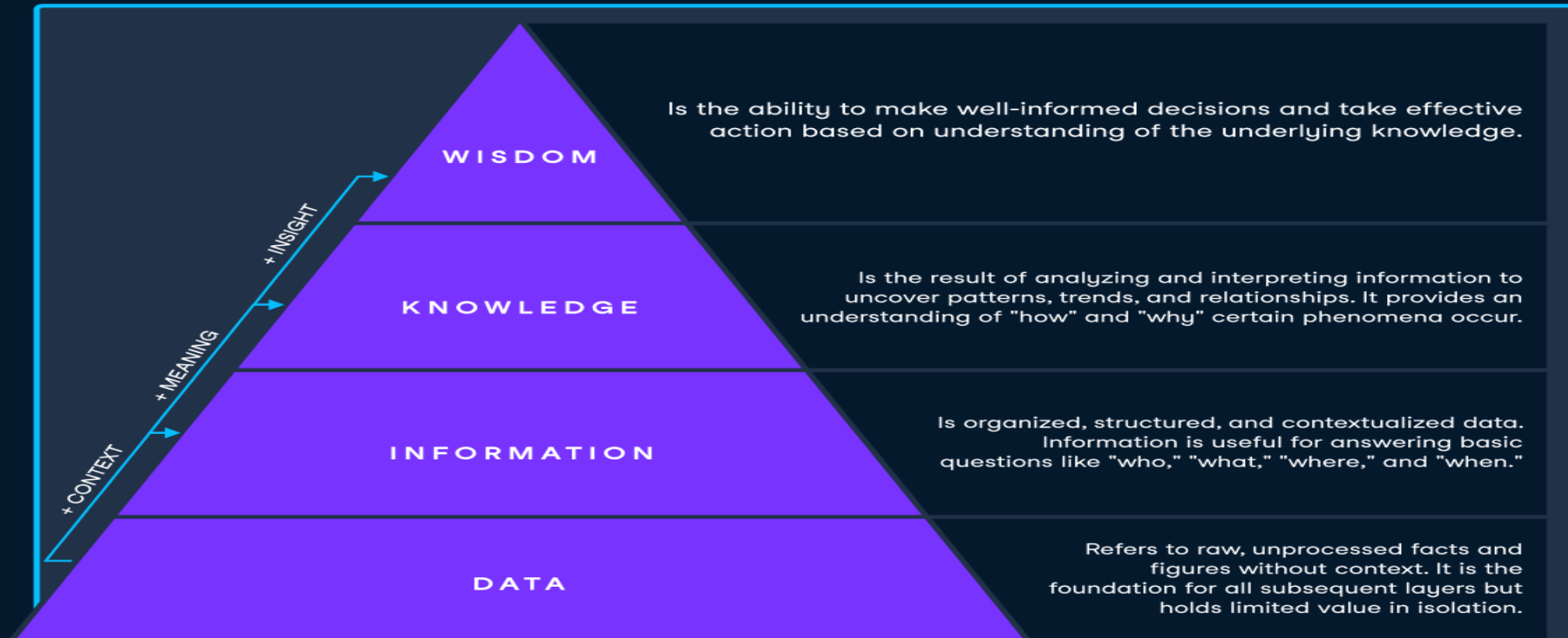
Function	Description	Key characteristics
Planning	Setting goals and developing strategies to achieve them	Future-oriented, involves decision-making and resource allocation
Organizing	Structuring the organization and assigning tasks	Focuses on creating a framework for coordinated action
Leading	Motivating and inspiring employees	Involves influencing and empowering people
Controlling	Monitoring progress and taking corrective actions	Ensures alignment with goals and objectives

- ▶ Information, knowledge, and wisdom are often used interchangeably.
- ▶ **Data** is the most basic level and refers to raw, unorganized facts and figures.
- ▶ For example, the height, weight, and age of a person are data points.
- ▶ **Information** is organized and processed data that has meaning and context.
- ▶ **Knowledge** is the application of information to solve problems and make decisions.
- ▶ For example, knowing about a person's height, weight, and age could be used to determine their body mass index (BMI), which is a measure of body fatness.
- ▶ This knowledge could then be used to provide the person with personalized health advice.
- ▶ **Wisdom** is the deepest level of understanding and insight gained from experience and knowledge.

Data-Information-Knowledge-Wisdom Pyramid



The Data-Information-Knowledge-Wisdom (DIKW) pyramid illustrates the progression of raw data to valuable insights. It gives you a framework to discuss the level of meaning and utility within data. Each level of the pyramid builds on lower levels, and to effectively make data-driven decisions, you need all four levels.



DATA, INFORMATION, KNOWLEDGE AND WISDOM

- ▶ As we move up the pyramid, the value of the data increases.
- ▶ Data is essential for creating information,
- ▶ Information is necessary for generating knowledge, and
- ▶ Knowledge is crucial for developing wisdom.

Concept	Definition	Characteristics
Data	Raw, unorganized facts and figures	Meaningless and valueless in isolation
Information	Organized and processed data with meaning and context	Useful for understanding and making decisions
Knowledge	Application of information to solve problems and make decisions	Valuable for solving specific problems and making informed choices
Wisdom	Deep understanding and insight gained from experience and knowledge	Enables sound judgment, ethical decision-making, and long-term perspective

- ▶ Managers need a variety of information to make informed decisions, plan for the future, and solve problems.
- ▶ The specific information needs of managers will vary depending on their level and area of responsibility.
- ▶ **However, some of the common information needs of managers include:**
- ▶ **Industry and market information:** Obtained from a variety of sources, such as accounting statements, financial reports, and financial analysis tools.
- ▶ **Customer information:** can be obtained from a variety of sources, such as customer surveys, market research reports, and customer feedback.

- ▶ **Operational information:** can be obtained from a variety of sources, such as production reports, inventory data, and quality control reports.
- ▶ Managers can access information from a variety of sources, both internal and external.
- ▶ **Internal sources:** company documents, reports, and data, as well as the knowledge and expertise of employees.
- ▶ **External sources:** industry publications, trade shows, government statistics, and market research reports.

- ▶ A structured and organized approach to understanding, designing, implementing, and managing IS within an organization.
- ▶ A framework for information systems provides a systematic way to plan, analyze, design, implement, and manage information systems throughout their lifecycle.
- ▶ Here is a general framework that encompasses key stages in the information systems lifecycle:
- ▶ **Step1:Understanding Business Objectives:**
 - **Business Strategy Alignment:** Align information systems strategy with overall business strategy to ensure that technology initiatives support organizational goals.

▶ Step 2: **Strategic Planning:**

- **SWOT Analysis:** Conduct a thorough analysis of Strengths, Weaknesses, Opportunities, and Threats to identify strategic directions.
- **Goal Setting:** Define clear and measurable goals that the information system should achieve.

▶ Step 3: **Requirements Analysis:**

- **Stakeholder Identification:** Identify and involve key stakeholders to understand their needs and expectations.
- **Feasibility Study:** Evaluate the technical, operational, and financial feasibility of the proposed information system.

► Step 4: System Design:

- **Architecture Design:** Define the overall structure and architecture of the information system.
- **Database Design:** Develop the database schema and data model.
- **Interface Design:** Design user interfaces and interactions.
- **Network Design:** Plan the network infrastructure required for the system.

► Step 5: Development and Implementation:

- **Coding:** Write, test, and debug the software code.
- **Integration:** Integrate various components and modules of the system.
- **Testing:** Conduct thorough testing to ensure the system meets requirements and is free of defects.
- **Deployment:** Deploy the information system in the production environment.

A FRAMEWORK FOR INFORMATION SYSTEMS

▶ **Step 6: Training and User Acceptance:**

- **Training Programs:** Provide training to users on how to use the new system.
- **User Acceptance Testing:** Validate that the system meets user expectations and requirements.

▶ **Step 7: Operation and Maintenance:**

- **System Monitoring:** Implement monitoring tools to track system performance.
- **Bug Fixes and Updates:** Address and fix any issues that arise and release updates as needed.
- **Security Management:** Implement security measures to protect the system from threats.

A FRAMEWORK FOR INFORMATION SYSTEMS

▶ Step 8 :Evaluation and Enhancement:

- **Performance Evaluation:** Assess the performance of the information system against predefined metrics.
- **Feedback and Improvement:** Gather feedback from users and stakeholders to identify areas for improvement.
- **Continuous Enhancement:** Implement changes and updates to enhance the system based on feedback and changing requirements.

▶ Step 9: Documentation:

- **User Manuals:** Provide comprehensive documentation for users.
- **Technical Documentation:** Document the technical aspects of the system for maintenance and future development.

▶ Step 10:Governance and Compliance:

- **Compliance Management:** Ensure that the information system complies with relevant laws, regulations, and industry standards.
- **Governance Structure:** Establish governance mechanisms to oversee and manage the information system.

A FRAMEWORK FOR INFORMATION SYSTEMS

- ▶ This framework is iterative and emphasizes the importance of ongoing evaluation, feedback, and improvement throughout the information system's lifecycle.
- ▶ Organizations may adapt and customize this framework to suit their unique needs, industry requirements, and technological landscape.

Business system vs E-business system

30

Aspect	Business Systems	E-Business Systems
Scope	▪ Broader perspective covering various business functions and processes.	▪ Specific focus on electronic transactions and digital interactions.
Functionality	▪ Encompasses a wide range of business operations, both electronic and manual.	▪ Primarily focused on online transactions, digital communication, and e-commerce.
Technology Usage	▪ Utilizes a mix of technologies, including both electronic and manual processes.	▪ Heavily relies on digital technologies, the internet, and electronic communication.
Automation Level	▪ Automation levels can vary across different processes within the system.	▪ Tends to have higher levels of automation, especially in online transactions and processes.
Examples	▪ Enterprise Resource Planning (ERP) systems.	▪ E-commerce platforms, Customer Relationship Management (CRM) systems.
Integration	▪ Integrates various business processes and functions across an organization.	▪ Often integrated into broader business systems, contributing to overall efficiency.

Chapter Three: *Information Technology*

Hardware: Computing, Storing and communicating

- ▶ In the context of information systems, hardware plays a crucial role in computing, storing, and communicating data.
- ▶ **Computing Hardware:**
 - Computing hardware refers to the physical components that make up a computer system and enable it to perform computational tasks.
 - **Role in Information Systems:** Computing hardware is essential for running software applications, processing data, and executing the functions of information systems.

► Key Components:

- **Central Processing Unit (CPU):** The "brain" of the computer, responsible for executing instructions and performing calculations.
- **Memory (RAM):** Provides temporary storage for data and program instructions that the CPU is actively using.
- **Storage Devices:** Hard drives (HDD), solid-state drives (SSD), or other storage media for long-term data retention.
- **Motherboard:** The main circuit board connecting and facilitating communication between various hardware components.
- **Peripheral Devices:** Input devices (e.g., keyboard, mouse) and output devices (e.g., monitor, printer) for user interaction.

► Storing Hardware:

- Storing hardware involves devices and components that facilitate the storage and retrieval of data.
- **Role in Information Systems:** Storing hardware is critical for preserving data, files, and applications. It supports data management, retrieval, and backup processes within information systems.

► Key Components:

- **Hard Disk Drives (HDD) and Solid-State Drives (SSD):** Main storage devices for saving data over the long term.
- **Network Attached Storage (NAS):** Storage devices connected to a network, allowing multiple users to access shared data.
- **Cloud Storage:** Remote servers providing scalable and accessible storage over the internet.

► **Communicating Hardware:**

- Communicating hardware involves devices and infrastructure that enable the transmission of data and communication within and between systems.
- **Role in Information Systems:** Communicating hardware is essential for establishing networks, enabling collaboration, and supporting the flow of information within and between organizations.

► **Key Components:**

- **Network Devices:** Routers, switches, and hubs that facilitate data transfer between devices on a network.
- **Communication Protocols:** Standardized rules for data exchange, ensuring compatibility and reliability (e.g., TCP/IP for the internet).
- **Network Cabling and Wireless Technologies:** Physical cables or wireless connections for transmitting data.

► Integration of Hardware in Information Systems:

- **Interdependence:** Computing, storing, and communicating hardware components work together to create a cohesive information system.
- **Scalability:** Systems can be designed with scalability in mind, allowing for the expansion or modification of hardware components to accommodate growing data and user requirements.
- **Reliability and Redundancy:** Redundancy measures, such as backup systems and fault-tolerant hardware configurations, ensure system reliability and minimize disruptions.
- **Security Considerations:** Hardware plays a role in implementing security measures, such as firewalls, encryption, and access controls, to protect data and ensure system integrity.

Hardware: Computing, Storing and communicating

- ▶ A well-designed information system considers the integration and optimization of hardware components to meet the organization's computing, storage, and communication needs efficiently and effectively.

The Three P's in IT

20

- ▶ The three key components often referred to as the "Three P's" are ***Program, People, and Procedure***.
- ▶ These elements are critical for the effective functioning and success of information systems. Let's explore each component:
- ▶ **Program:**
 - a program refers to the software applications and computer programs that are designed to perform specific tasks or functions.
- ▶ **Role in Information Technology:**
 - Programs are the functional entities that execute tasks, process data, and enable the use of computing resources.
 - They can include applications for data processing, communication, security, and various business functions.

► Key Aspects:

- **Software Applications:** Programs include a wide range of software applications, from operating systems to specialized business applications.
- **Coding and Algorithms:** Programs are written using programming languages and algorithms to instruct computers on how to perform tasks.
- **Integration:** Programs often work together within an information system, interacting to achieve overall objectives.

The Three P's in IT

▶ **People:**

- ▶ The "People" component refers to the individuals involved in the development, management, and use of information technology within an organization.

▶ **Role in Information Technology:**

- People are critical for the design, development, and operation of information systems.
- Effective communication and collaboration among people are essential for successful IT implementation.
- End users play a crucial role in the adoption and usability of IT systems.

► Key Aspects:

- **IT Professionals:** Developers, system administrators, network engineers, and other specialists involved in designing, implementing, and maintaining IT systems.
- **End Users:** Individuals who interact with and use the information systems for their daily tasks.
- **Management and Decision-Makers:** Leaders who make strategic decisions about IT investments, policies, and overall direction.

The Three P's in IT

12

▶ **Procedure:**

- ▶ Procedures refer to the documented processes, methods, and guidelines that dictate how tasks are to be performed within an organization.

▶ **Role in Information Technology:**

- Procedures ensure consistency, efficiency, and adherence to best practices in the use and management of IT resources.
- They contribute to the organization's compliance with regulations, security standards, and quality assurance.

▶ **Key Aspects:**

- **Standard Operating Procedures (SOPs):** Documented step-by-step guides for performing specific tasks or processes.
- **Security Protocols:** Procedures outlining security measures, access controls, and data protection practices.
- **Change Management Processes:** Procedures for implementing changes to IT systems in a controlled and organized manner.

► **Integration of the Three P's:**

- **Holistic Approach:** Success in information technology requires a holistic approach that considers the interdependence of programs, people, and procedures.
- **Training and Education:** People need to be trained to effectively use and manage IT systems. This includes end-user training and ongoing professional development for IT professionals.
- **Alignment with Business Objectives:** Procedures should align with the overall business objectives of the organization, and programs should support the processes outlined in the procedures.
- **Continuous Improvement:** Regular evaluation and improvement of programs, training methods, and procedures are essential to adapt to changing technology and business requirements.

Communication Technologies

- ▶ Communication technologies refer to the various tools, systems, and methods used to transmit, exchange, and receive information between individuals or entities.
- ▶ These technologies play a crucial role in connecting people, organizations, and devices.
- ▶ Here are some key communication technologies:

1. Internet:

- **Definition:** The global system of interconnected computer networks that use the Internet Protocol Suite (TCP/IP) to link devices worldwide.
- **Applications:** Email, web browsing, social media, online collaboration, and various internet-based services.

2. Email:

- **Definition:** Electronic mail allows the exchange of messages and files between users with an email address.
- **Applications:** Formal and informal communication, document sharing, and collaboration.

3. Instant Messaging (IM) and Chat:

- **Definition:** Real-time text-based communication platforms that allow users to chat with each other.
- **Applications:** Workplace communication, personal messaging, and group collaboration.

▶ Voice over Internet Protocol (VoIP):

- **Definition:** Technology that enables voice communication and multimedia sessions over the internet, bypassing traditional telephone networks.
- **Applications:** Internet calling, video conferencing, and online meetings.

► Video Conferencing:

- **Definition:** Real-time communication that allows participants in different locations to see and hear each other.
- **Applications:** Business meetings, remote collaboration, and virtual classrooms.

► Unified Communications (UC):

- **Definition:** Integration of various communication tools and platforms into a single interface, enhancing collaboration and productivity.
- **Applications:** Unified messaging, presence information, and integrated voice and video communication.

► Social Media:

- **Definition:** Online platforms that facilitate social interaction, content sharing, and networking.
- **Applications:** Social networking, content sharing, and business promotion.

► Mobile Communication:

- **Definition:** Wireless communication technologies that enable voice and data transmission via mobile devices.
- **Applications:** Mobile calls, text messaging, mobile internet, and mobile apps.

► Wireless Technologies:

- **Wi-Fi:** Wireless networking technology that allows devices to connect to the internet without physical cables.
- **Bluetooth:** Short-range wireless technology for connecting devices such as headphones, keyboards, and smartphones.

▶ **Satellite Communication:**

- **Definition:** Communication using satellites to transmit signals over long distances.
- **Applications:** Global communication, broadcasting, and remote sensing.

▶ **Radio Frequency Identification (RFID):**

- **Definition:** Technology that uses wireless communication to identify and track objects using RFID tags.
- **Applications:** Inventory management, asset tracking, and access control.

▶ **NFC (Near Field Communication):**

- **Definition:** Short-range wireless technology that enables communication between devices in close proximity.
- **Applications:** Contactless payments, mobile device pairing, and data transfer.

► Web Conferencing:

- **Definition:** Online meetings that allow participants to share presentations, documents, and applications in real-time.
- **Applications:** Virtual meetings, collaborative work sessions, and remote training.

► Cable and Fiber Optic Communication:

- **Definition:** High-speed data transmission using cables or fiber optics.
- **Applications:** Broadband internet, cable television, and high-speed data networks.

- ▶ It involves the organization, storage, retrieval, and protection of data in a database.
- ▶ Databases are crucial components in many software applications and systems, providing a structured way to store and manage information.
- ▶ Here are some key aspects of database management:

A. Database Design:

- i. Entity-Relationship (ER) Modeling:** Designing the structure of the database by defining entities, their attributes, and the relationships between them.
- ii. Normalization:** Ensuring that the database design is optimized and free from redundancy by applying normalization techniques.

B. Database Types:

- I. **Relational Databases:** Use tables to store data, with relationships defined between tables.
- II. **NoSQL Databases:** Provide flexible schema designs and are suitable for handling unstructured or semi-structured data.
- III. **Graph Databases:** Designed for managing data with complex relationships.

C. Database Languages:

- I. **SQL (Structured Query Language):** Standard language for interacting with relational databases, used for querying, updating, and managing data.
- II. **NoSQL Query Languages:** Each NoSQL database type may have its own query language.

D. Database Operations:

- I. **CRUD Operations:** Create, Read, Update, and Delete operations for managing data.
- II. **Transactions:** Ensuring data consistency by grouping operations into atomic transactions.

E. Data Integrity and Constraints:

- I. **Primary Keys and Foreign Keys:** Ensuring uniqueness and relationships between tables.
- II. **Check Constraints:** Enforcing specific conditions on data values.

F. Indexing and Performance Tuning:

- I. **Indexes:** Improving query performance by creating indexes on columns frequently used in search conditions.
- II. **Query Optimization:** Tuning queries for better performance.

G. Backup and Recovery:

- I. **Regular Backups:** Creating backups of the database to prevent data loss.
- II. **Recovery Planning:** Establishing procedures for recovering data in case of failures.

H. Security:

- I. **Access Control:** Managing user permissions to ensure data security.
- II. **Encryption:** Protecting sensitive data through encryption techniques.

I. Scalability and Replication:

- I. **Scaling Strategies:** Ensuring that the database can handle increased loads.
- II. **Replication:** Creating copies of the database for fault tolerance and improved performance.

J. Database Administration:

- I. **Monitoring and Maintenance:** Regularly monitoring database performance and performing maintenance tasks.
- II. **User Management:** Managing user accounts, roles, and permissions.

K. Cloud Databases:

- I. **Cloud-Based Solutions:** Utilizing database services provided by cloud providers for scalability and accessibility.